SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Dr.V.Parthipan

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I **P. Harsha Vardhan (192411179)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **p harsha**

ANSWERS TO DEBUGGING & OPTIMISATION TASK

Question 1: Smart Office WLAN/WWAN Connectivity Issues

Background:

The company relies on two distinct wireless technologies. The Wireless Local Area Network (WLAN, IEEE 802.11 Wi-Fi) provides in-building connectivity for laptops, IP phones, printers, and IoT devices through access points spread across six floors. The Wireless Wide Area Network (WWAN, 4G/5G cellular) provides connectivity for field engineers and sales staff outside the building. Both networks fail differently — WLAN issues stem from in-building RF planning and capacity, while WWAN issues stem from carrier coverage and mobility — so each must be diagnosed and optimized separately even though both share the same overall goal of seamless connectivity.

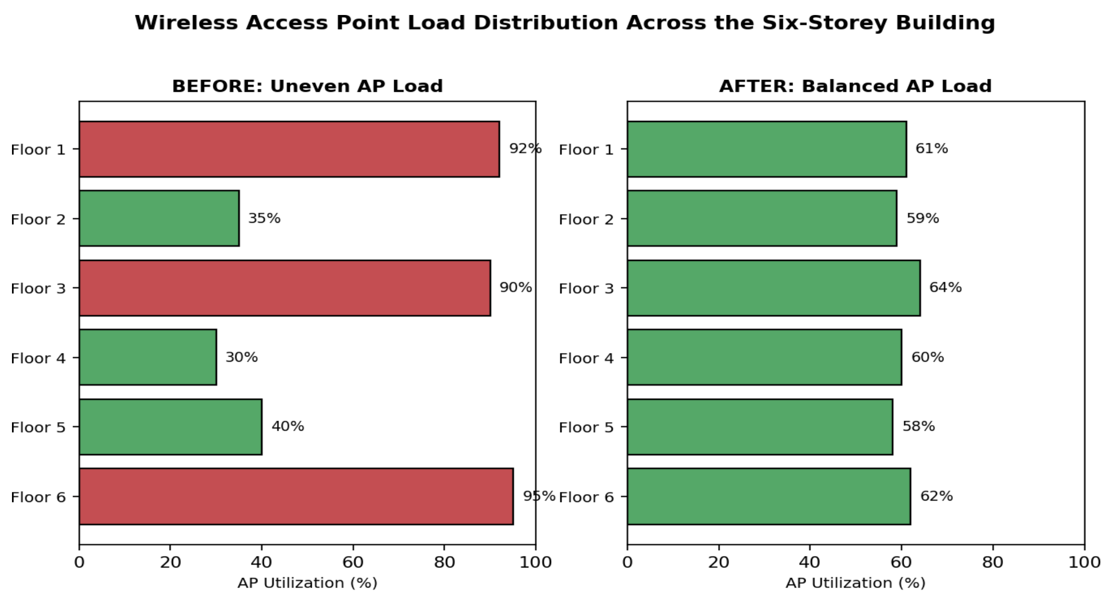
1. Problem Identification (Root Causes):

Slow speeds & disconnections during peak hours: too many devices (1,200+ employees, laptops, IP phones, IoT devices) associating with a limited number of access points (APs), causing airtime contention. Frequent disconnections while moving between floors: absence of fast roaming standards (802.11r/k/v), so devices drop and re-associate instead of transitioning seamlessly. Weak signals in conference rooms: poor AP placement/transmit-power planning and building material attenuation across a six-storey structure. Video conferencing freezing: bandwidth congestion and jitter caused by lack of QoS prioritization for real-time traffic. Uneven AP utilization: static AP configuration with no load balancing, so some APs are overloaded while neighbouring APs sit idle. WWAN loss for field engineers: coverage gaps in rural/highway areas and no failover between carriers. Outdated security protocols: legacy WEP/WPA-TKIP devices still permitted on the network, exposing the company to attacks such as packet sniffing and key-cracking.

2. Debugging (Systematic Diagnosis):

Step 1 — Conduct a full wireless site survey / heat-map analysis across all six floors to locate weak-signal and dead zones. Step 2 — Use the Wireless LAN Controller (WLC) dashboard to review per-AP client counts and channel utilization, confirming the overloaded vs. underutilized APs. Step 3 — Run a spectrum analysis to check for channel overlap/co-channel interference, particularly in conference rooms. Step 4 — Inspect roaming logs to confirm whether 802.11r/k/v fast-transition features are enabled on the APs and clients. Step 5 — Correlate field-engineer WWAN drop reports with carrier coverage maps to identify genuine coverage gaps versus device/SIM configuration issues. Step 6 — Audit connected-device security logs (RADIUS/AAA server) to list all devices still authenticating with WEP or WPA-TKIP.

Before / After AP Load Distribution:



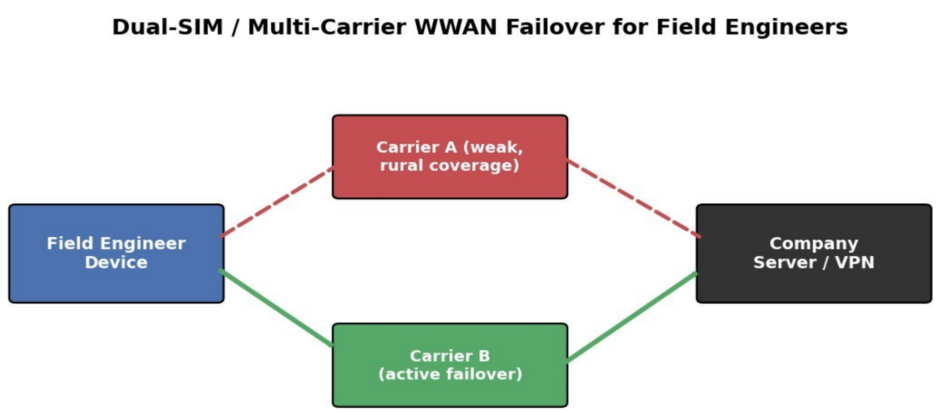
3. Optimization (Recommended Fixes):

Redesign the AP layout using the site-survey results; adjust transmit power and assign non-overlapping channels (1, 6, 11 on 2.4 GHz; non-overlapping 20/40 MHz channels on 5 GHz) to reduce interference. Enable band steering and client load balancing on the WLC so devices are pushed toward the least-congested AP and, where possible, toward the 5 GHz band. Enable 802.11r (fast BSS transition), 802.11k (neighbour reports) and 802.11v (BSS transition management) so devices roam between floors without dropping the connection. Apply Wi-Fi Multimedia (WMM) QoS with DSCP marking so video-conferencing and VoIP packets are prioritized ahead of bulk file transfers. For WWAN, provision dual-SIM/multi-carrier modems with automatic failover and, where coverage is weak, deploy signal boosters or satellite backup links for field engineers in rural/highway areas. Enforce WPA3-Enterprise (or at minimum WPA2-Enterprise with 802.1X) across the entire WLAN, and use Network Access Control (NAC) to automatically quarantine any device still attempting to connect using WEP or WPA-TKIP until it is reconfigured.

802.11 Fast-Roaming Standards Enabled:

Standard	Function	Benefit for Floor-to-Floor Roaming
802.11r	Fast BSS Transition (FT)	Pre-negotiates security keys with the target AP so re-authentication is near-instant
802.11k	Radio Resource Management	Client receives a neighbour-AP report, so it knows the best AP to roam to
802.11v	BSS Transition Management	Network can proactively steer a client to a better AP before signal fully degrades

WWAN Failover for Field Engineers:



Security Protocol Comparison:

Protocol	Encryption	Status
WEP	RC4 (weak, static key)	Deprecated — crackable within minutes; must be disabled
WPA / WPA-TKIP	RC4-based TKIP	Legacy — vulnerable to known key-recovery attacks; phase out
WPA2-Enterprise (802.1X)	AES-CCMP	Minimum acceptable baseline for corporate WLAN
WPA3-Enterprise	AES-GCMP, SAE handshake	Recommended target state — resistant to offline dictionary attacks

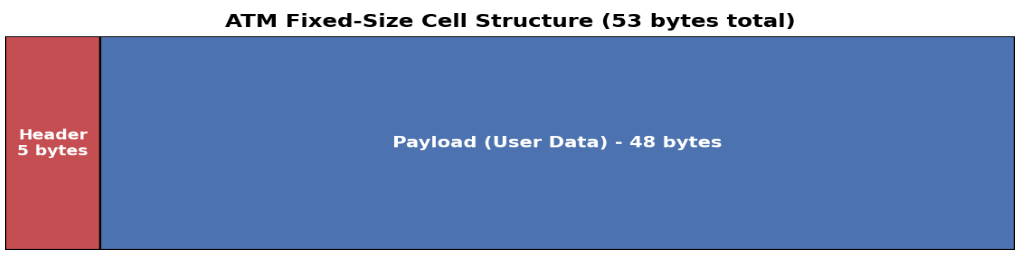
4. Analysis:

With 1,200+ users spread over six floors, capacity planning should target roughly 25–30 concurrent clients per AP for acceptable performance, meaning AP density and placement — not just AP count — determines quality of service. Moving from static, unmanaged AP behaviour to controller-based band steering and fast roaming directly addresses the uneven-load and floor-transition symptoms, while DSCP-based QoS ensures that even under peak load, latency-sensitive video/voice traffic is not starved by bulk data transfers. Enforcing modern security protocols closes the vulnerability window without requiring a hardware refresh, since most modern devices already support WPA2/WPA3, and the WWAN failover design ensures field engineers are not left disconnected simply because one carrier has a coverage gap in a given rural stretch.

Question 2: ATM Network Performance Issues

Background:

Asynchronous Transfer Mode (ATM) transmits all traffic in fixed-size 53-byte cells (5-byte header + 48-byte payload), regardless of the type of data being carried. This fixed cell size is what allows ATM to provide predictable, low-jitter switching suitable for voice and video, at the cost of some header overhead compared to variable-length packet formats.



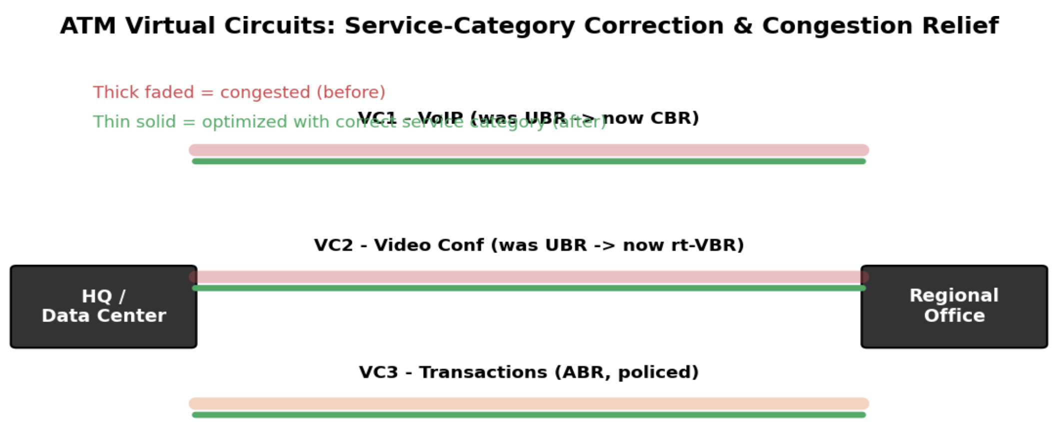
1. Problem Identification (Root Causes):

Delays in online transactions, jittery voice calls and degraded video conferencing point to Virtual Circuits (VCs) that are congested and/or assigned to the wrong ATM service category. ATM defines five service categories — CBR (Constant Bit Rate), rt-VBR (real-time Variable Bit Rate), nrt-VBR (non-real-time VBR), ABR (Available Bit Rate) and UBR (Unspecified Bit Rate) — each with different bandwidth guarantees and delay/jitter tolerances. If VoIP or video conferencing is running over UBR (best-effort, no guarantees) instead of CBR/rt-VBR, jitter and cell loss are expected. High switch utilization indicates bandwidth is not being efficiently allocated among traffic classes, and congestion on some VCs while others remain unused suggests static VC routing rather than dynamic, load-aware path selection.

2. Debugging (Systematic Diagnosis):

Step 1 — Monitor Cell Loss Ratio (CLR) and Cell Delay Variation (CDV) per VC at each switch to quantify which VCs are actually congested. Step 2 — Cross-check the ATM service category assigned to each application (banking transactions, VoIP, video conferencing) against its actual traffic profile and QoS requirement. Step 3 — Examine switch-level utilization statistics to identify which switches are operating near capacity and whether load is unevenly distributed across available trunks. Step 4 — Verify bandwidth reservation (Peak Cell Rate / Sustainable Cell Rate) for each VC against the negotiated traffic contract to check for over-subscription. Step 5 — Review the PNNI (Private Network-to-Network Interface) routing configuration to see whether VC paths are chosen dynamically or statically.

VC Congestion and Service-Category Correction:



ATM Service Categories and Suitable Applications:

Category	QoS Guarantee	Best Suited For
CBR (Constant Bit Rate)	Guaranteed bandwidth, tightest delay/CDV bound	VoIP, circuit emulation
rt-VBR	Guaranteed average + burst rate, bounded delay	Real-time video conferencing
nrt-VBR	Guaranteed average rate, no strict delay bound	Transaction processing, database queries
ABR (Available Bit Rate)	Minimum rate + feedback-based fair sharing	Bursty banking transactions
UBR (Unspecified Bit Rate)	Best-effort, no guarantees	Bulk file transfer, non-critical data

3. Optimization (Recommended Fixes):

Reassign VoIP calls and video conferencing sessions from UBR/ABR to CBR or rt-VBR, which guarantee bandwidth and bound cell delay variation, directly reducing jitter and packet loss. Keep bursty, delay-tolerant banking-transaction traffic on ABR with proper flow-control feedback (Resource Management cells) so it can use spare bandwidth without starving real-time traffic. Apply traffic shaping and policing at the User-Network Interface (UNI) to prevent any single VC from exceeding its negotiated contract and congesting the switch. Increase trunk bandwidth or add parallel VCs on links identified as chronically congested, and enable PNNI-based dynamic routing so new VCs are automatically routed via the least-congested path. Introduce Connection Admission Control (CAC) so that new VC requests are only accepted if sufficient bandwidth is available, preventing overload from occurring in the first place.

4. Analysis:

ATM's strength is guaranteed QoS through the traffic-contract model — the issues described are symptomatic of that model being misused (wrong service category) rather than of any inherent weakness in ATM itself. Correcting the service-category assignment (CBR/rt-VBR for real-time traffic) is the single highest-impact fix, since it directly targets the jitter and packet-loss symptoms reported by users, while CAC and PNNI-based dynamic routing prevent the same congestion pattern from recurring as traffic grows.

Question 3: MPLS Network Performance Issues

Background:

MPLS forwards packets based on short, fixed-length labels rather than performing a full IP lookup at every hop. Each Label Switch Router (LSR) maintains a Label Forwarding Information Base (LFIB) that maps an incoming label to an outgoing label and interface — a simple swap operation that is much faster than conventional routing. A Label Switched Path (LSP) is the specific end-to-end path a labelled packet follows through the backbone, and MPLS Traffic Engineering (MPLS-TE) is the mechanism that decides which LSP a given traffic flow should use, based on available bandwidth rather than only the shortest IGP path.

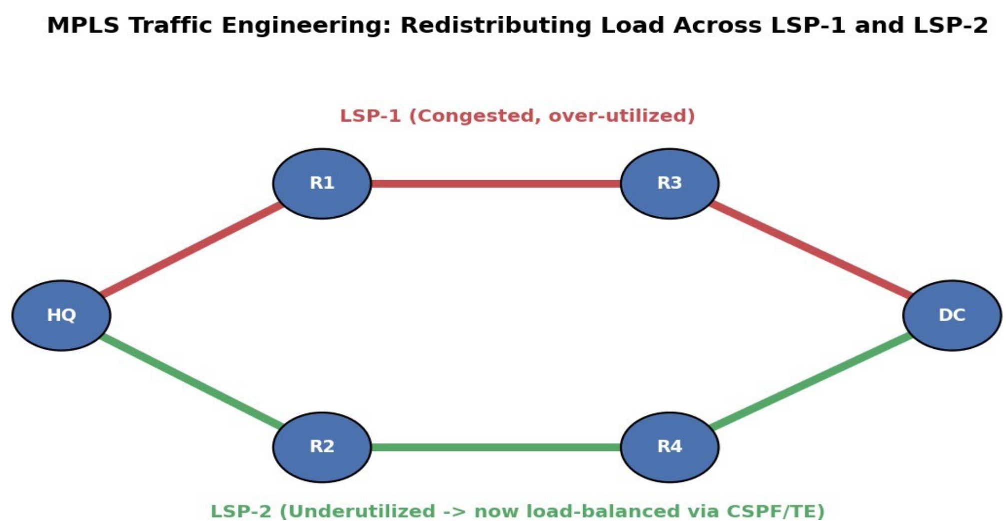
1. Problem Identification (Root Causes):

Slow access to cloud applications, delayed order processing, and poor voice/video quality point to Label Switched Paths (LSPs) that are congested while alternate paths across the same backbone remain underutilized. Incorrect label mappings on some routers mean packets can be forwarded along inefficient or even looping paths, increasing latency and, in the worst case, causing packet loss. The absence of properly configured traffic engineering (TE) means the network is relying on default IGP shortest-path routing rather than bandwidth-aware path selection, so traffic naturally piles up on the 'shortest' LSP even when a parallel LSP has spare capacity.

2. Debugging (Systematic Diagnosis):

Step 1 — Inspect the Label Forwarding Information Base (LFIB) on each Label Switch Router (LSR) to verify that incoming labels are being swapped to the correct outgoing label/interface. Step 2 — Use LSP ping and LSP traceroute (MPLS OAM tools) to confirm the actual forwarding path matches the intended path and to detect any misrouted or looping LSPs. Step 3 — Monitor per-LSP utilization and bandwidth-reservation statistics via the MPLS-TE tunnel database to identify which LSPs are congested and which have spare capacity. Step 4 — Review the CSPF (Constrained Shortest Path First) computation logs to check whether TE tunnels are being placed based on available bandwidth or purely on IGP metric. Step 5 — Check DiffServ/EXP-bit marking on real-time traffic (VoIP, video) to confirm it is being classified for priority treatment.

Traffic Engineering – Load Redistribution Across LSPs:



3. Optimization (Recommended Fixes):

Correct the faulty label bindings by resynchronizing LDP/RSVP-TE sessions on the affected routers and validating LFIB entries end-to-end. Enable MPLS Traffic Engineering with CSPF so that new LSPs are placed according to available bandwidth rather than only the IGP shortest path, automatically spreading load across LSP-1 and LSP-2 instead of concentrating it on one. Configure Fast Reroute (FRR) with pre-established backup LSPs so that if the primary LSP fails or becomes congested, traffic switches over with minimal disruption. Apply DiffServ-aware Traffic Engineering (DS-TE) so VoIP and video conferencing (EF/AF class) are guaranteed bandwidth on the least-congested LSP, while bulk ERP/inventory traffic uses best-effort LSPs. Where multiple equal-cost LSPs exist, enable load balancing (e.g., ECMP over TE tunnels) to use all available paths simultaneously rather than only the primary one.

Key MPLS Traffic-Engineering Mechanisms:

Mechanism	Purpose
CSPF (Constrained Shortest Path First)	Computes LSP paths based on bandwidth availability, not just IGP metric
RSVP-TE	Reserves bandwidth along the chosen LSP and signals label bindings hop-by-hop
Fast Reroute (FRR)	Pre-establishes a backup LSP so failover happens in milliseconds
DS-TE (DiffServ-aware TE)	Reserves separate bandwidth pools per traffic class (e.g., EF for voice) on each LSP

4. Analysis:

MPLS's core advantage — fast, deterministic forwarding via labels — is undermined when TE is not configured, because the network then behaves like a plain IGP-routed network that ignores available bandwidth. Enabling CSPF-based TE and DS-TE restores that advantage by making label-switched paths bandwidth- and priority-aware, which is the most direct fix for both the congestion imbalance and the QoS degradation reported by users.

Question 4: Warehouse Network – Error Detection & Correction

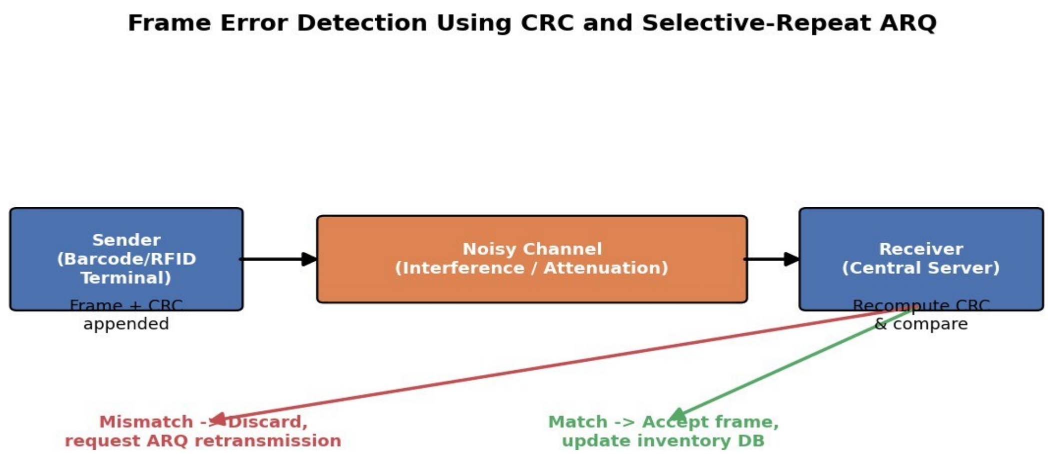
1. Problem Identification (Root Causes):

Incorrect shipment records, failed barcode scans, and inventory mismatches indicate that some corrupted frames are not being detected at all, while others are detected but only handled through retransmission, adding delay. The scenario states transmission errors are caused by electrical interference (from motors/robots in the warehouse), signal attenuation, and congestion during peak hours — all of which tend to introduce burst errors (multiple consecutive corrupted bits), a pattern that weak error-detection schemes such as a simple parity bit cannot reliably catch. The fact that 'some corrupted frames remain undetected' strongly suggests the current mechanism has an insufficient Hamming distance / weak checksum for the burst-error pattern typical of industrial electrical interference.

2. Debugging (Systematic Diagnosis):

Step 1 — Determine which error-detection method is currently in use (parity bit, simple checksum, or CRC) by inspecting the frame format at the data-link layer. Step 2 — Measure the Frame Error Rate (FER) during peak vs. off-peak hours to correlate error bursts with congestion and interference. Step 3 — Perform controlled fault injection (deliberately corrupt test frames) to measure the actual detection rate of the current mechanism versus a stronger alternative such as CRC-32. Step 4 — Review the retransmission (ARQ) scheme in use — Stop-and-Wait, Go-Back-N, or Selective-Repeat — to see whether undetected errors are compounded by an inefficient retransmission strategy. Step 5 — Audit the physical layer/cabling and RF environment around scanners and robots for sources of interference that can be shielded or relocated.

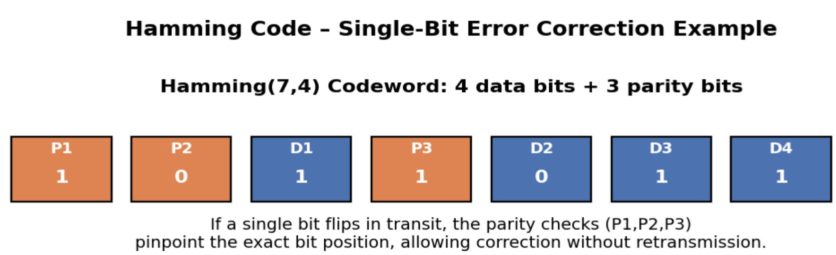
Improved Error Detection and Retransmission Flow:



Comparison of Error-Detection Techniques:

Technique	Detection Strength	Suitability for Warehouse Interference
Single Parity Bit	Detects only odd numbers of bit errors; misses burst errors	Weak — unsuitable, matches the reported 'undetected errors'
Simple Checksum	Better than parity but can miss certain error patterns	Moderate — still not reliable for burst noise
CRC-32	Detects virtually all burst errors up to the CRC length and most longer bursts	Strong — recommended replacement
Hamming Code (FEC)	Corrects single-bit errors without retransmission	Useful add-on for latency-sensitive updates

Forward Error Correction Example (Hamming Code):



3. Optimization (Recommended Fixes):

Replace weak parity/simple-checksum detection with CRC-32, which has a much larger Hamming distance and reliably detects burst errors of the length typically caused by electrical interference. For latency-sensitive updates (e.g., real-time inventory sync), add Forward Error Correction (FEC) using a Hamming code so that single-bit errors can be corrected at the receiver without needing a retransmission round-trip. Replace an inefficient ARQ scheme with Selective-Repeat ARQ, so that only the specific corrupted frame is retransmitted rather than the whole window (as in Go-Back-N), reducing wasted bandwidth during peak congestion. Shield or reroute cabling near motors/robots, and where feasible move to fiber-optic links for interference-prone segments. Add an application-layer integrity check (e.g., a hash of the full shipment record) so that even a frame that passes CRC but is logically corrupted before reaching the database can still be caught before committing bad data.

Sample CRC Verification Logic (Receiver Side):

```
// Pseudocode: CRC-32 based frame verification
function receiveFrame(frame):
    data         = frame.payload
    received_crc = frame.crc_field
    computed_crc = CRC32(data)           // recompute CRC over payload

    if computed_crc == received_crc:
        acceptFrame(data)                // update inventory DB
        sendACK(frame.seq_no)
    else:
        discardFrame(frame)
        sendSelectiveRepeatNACK(frame.seq_no)    // request re-send of this
frame only
```

4. Analysis:

The key diagnostic clue in this scenario is that errors are sometimes undetected, not merely frequent — this points specifically at detection-scheme weakness rather than link quality alone. Upgrading to CRC-32 (and FEC where latency matters) is therefore the highest-priority fix, and pairing it with Selective-Repeat ARQ ensures the correction mechanism is also bandwidth-efficient under the peak-hour congestion described.

<u>**Question 5: Online Education Platform – Flow Control Issues**</u>

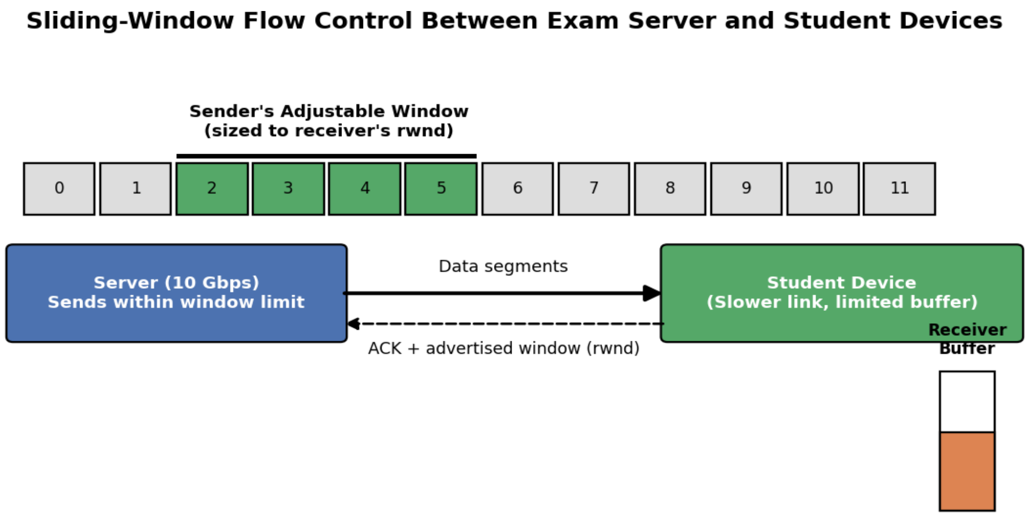
1. Problem Identification (Root Causes):

The server has a 10 Gbps connection while students use much slower broadband/mobile links — a classic sender/receiver speed mismatch. Symptoms described (incomplete downloads, duplicate packets, repeated retransmissions, receiver buffer overflow, application crashes) all point to flow control that is not adapting the sender's transmission rate to each receiver's actual processing/buffer capacity. If the sender keeps transmitting a fixed, large amount of data regardless of receiver feedback, the receiver's buffer fills faster than the device can process it, causing overflow, dropped packets, and — because the sender does not know this happened — unnecessary retransmissions and duplicates once timers expire.

2. Debugging (Systematic Diagnosis):

Step 1 — Capture and compare the receiver's advertised window (rwnd) values against the sender's actual outstanding (unacknowledged) data to see whether the sender is honouring the receiver's stated capacity. Step 2 — Check retransmission timeout (RTO) configuration — an RTO that is too short relative to the measured Round-Trip Time (RTT) will cause premature retransmissions and duplicate packets. Step 3 — Verify whether Selective Acknowledgment (SACK) is enabled; without it, a single lost segment can force retransmission of many already-received segments. Step 4 — Measure buffer occupancy on sample student devices during peak load to confirm overflow is actually occurring at the receiver rather than being dropped mid-network. Step 5 — Check whether the transport protocol in use provides flow control at all (e.g., raw UDP streaming without any windowing) versus TCP with proper window scaling.

Sliding-Window Flow Control Between Server and Students:



Flow-Control Mechanisms Compared:

Mechanism	How Rate Is Controlled	Weakness if Misconfigured
Stop-and-Wait	Sender waits for ACK before sending next frame	Very low throughput on high-latency links
Fixed-size Sliding Window	Sender may have a fixed number of unacknowledged frames outstanding	If window ignores receiver capacity, buffer overflow still occurs
Dynamic Sliding Window (rwnd-based)	Window size adjusts to the receiver's advertised buffer space	Requires receiver to send timely, accurate ACK/window updates
Congestion Control (CUBIC/BBR)	Sending rate adapts to estimated network bandwidth-delay product	Needs accurate RTT measurement; poor estimates cause under/over-sending

Worked Example — Why a Fixed Large Window Overflows a Slow Receiver:

If the server's window allows 100 KB of unacknowledged data in flight, but a mobile student's device can only buffer 20 KB before its application processes and clears data, the server will have overrun the buffer by 80 KB before the first ACK/window update is even received — explaining the observed buffer overflow and application crashes. Dynamically shrinking the window to match the receiver's advertised rwnd (here, 20 KB) prevents this overrun entirely.

3. Optimization (Recommended Fixes):

Implement (or correctly tune) sliding-window flow control so the sender's window size is dynamically bounded by the minimum of the congestion window (cwnd) and the receiver's advertised window (rwnd) — ensuring the server never sends faster than the slowest student's device/link can absorb. Enable TCP window scaling and Selective Acknowledgment (SACK) so that only genuinely lost segments are retransmitted, reducing duplicate traffic. Use a modern congestion-control algorithm (e.g., CUBIC or BBR) so the sending rate adapts smoothly to the measured bandwidth-delay product of each connection instead of overwhelming slower links. Offload static content (question papers, images, videos) to a Content Delivery Network (CDN) or multiple regional servers so the central 10 Gbps server is not the sole bottleneck for 15,000 concurrent students. Apply exponential backoff on retransmission timers to prevent duplicate-retransmission storms when many students experience timeouts simultaneously during peak load.

4. Analysis:

The core mismatch here — a very fast sender and many slow, heterogeneous receivers — is exactly the problem sliding-window flow control (with receiver-driven window sizing) is designed to solve; the symptoms described (buffer overflow, duplicates, poor throughput) are the textbook signature of a fixed-rate sender ignoring receiver feedback. Combining proper window-based flow control with CDN offloading addresses both the protocol-level cause and the underlying scale problem of one server serving 15,000 students simultaneously.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation